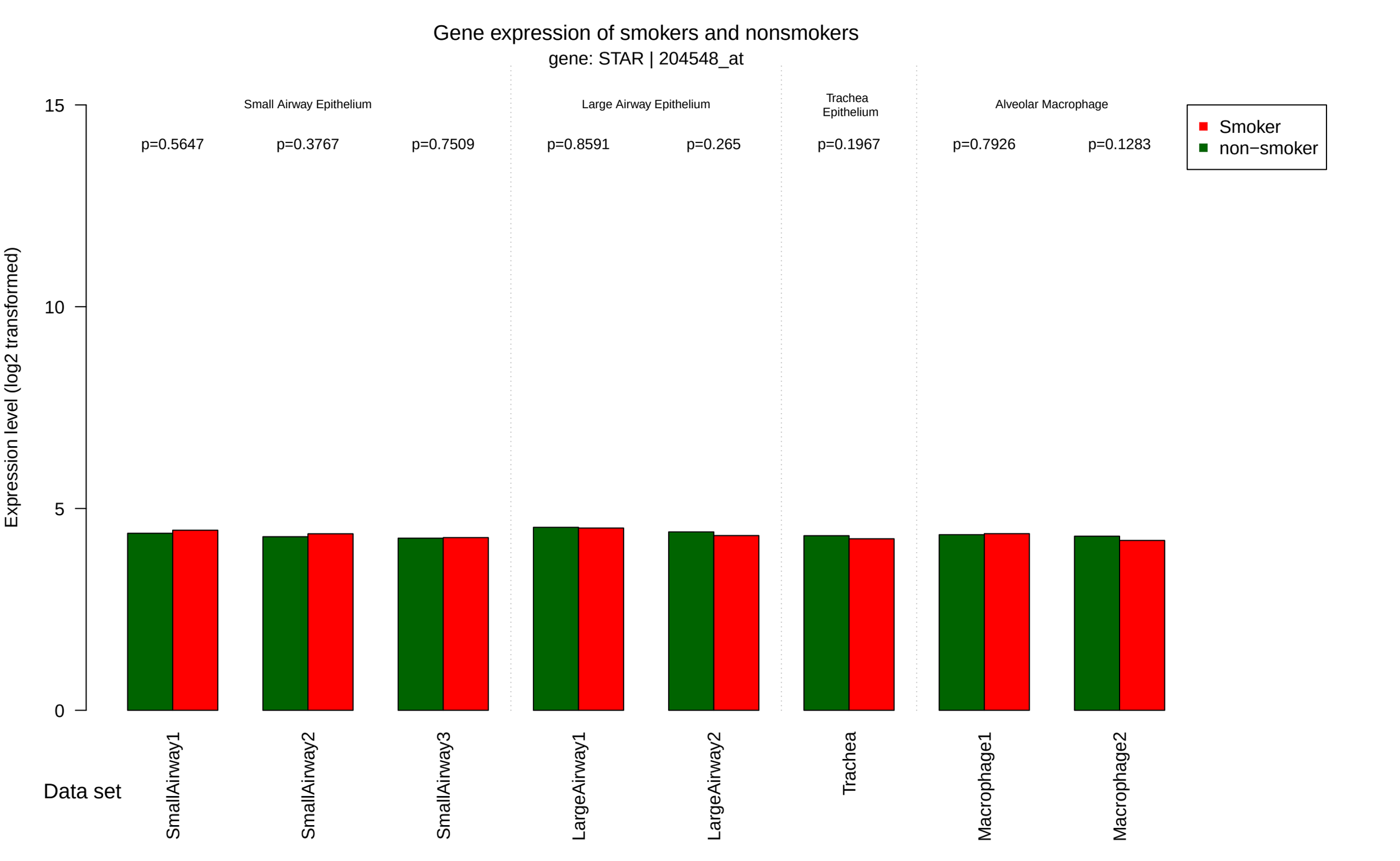
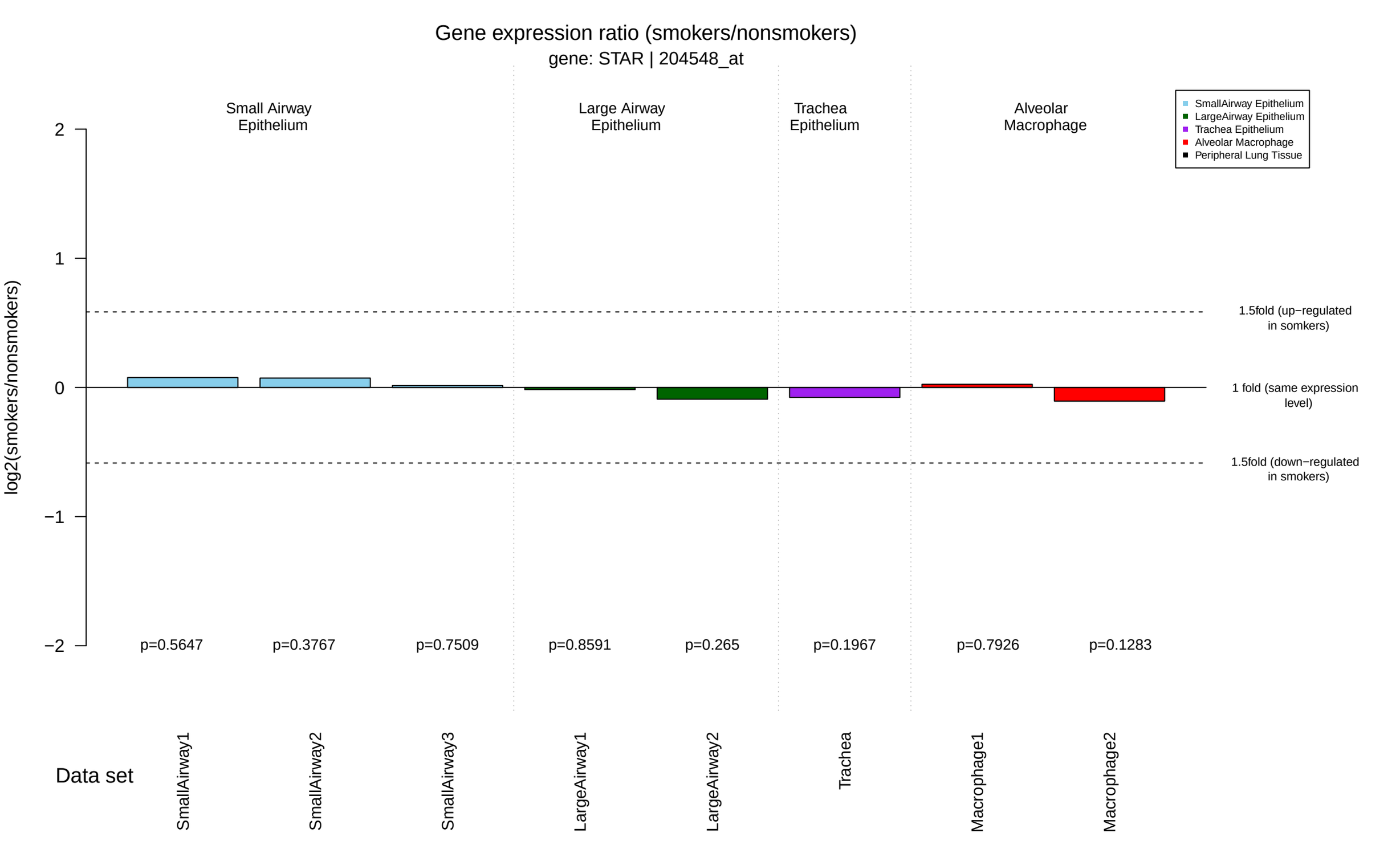
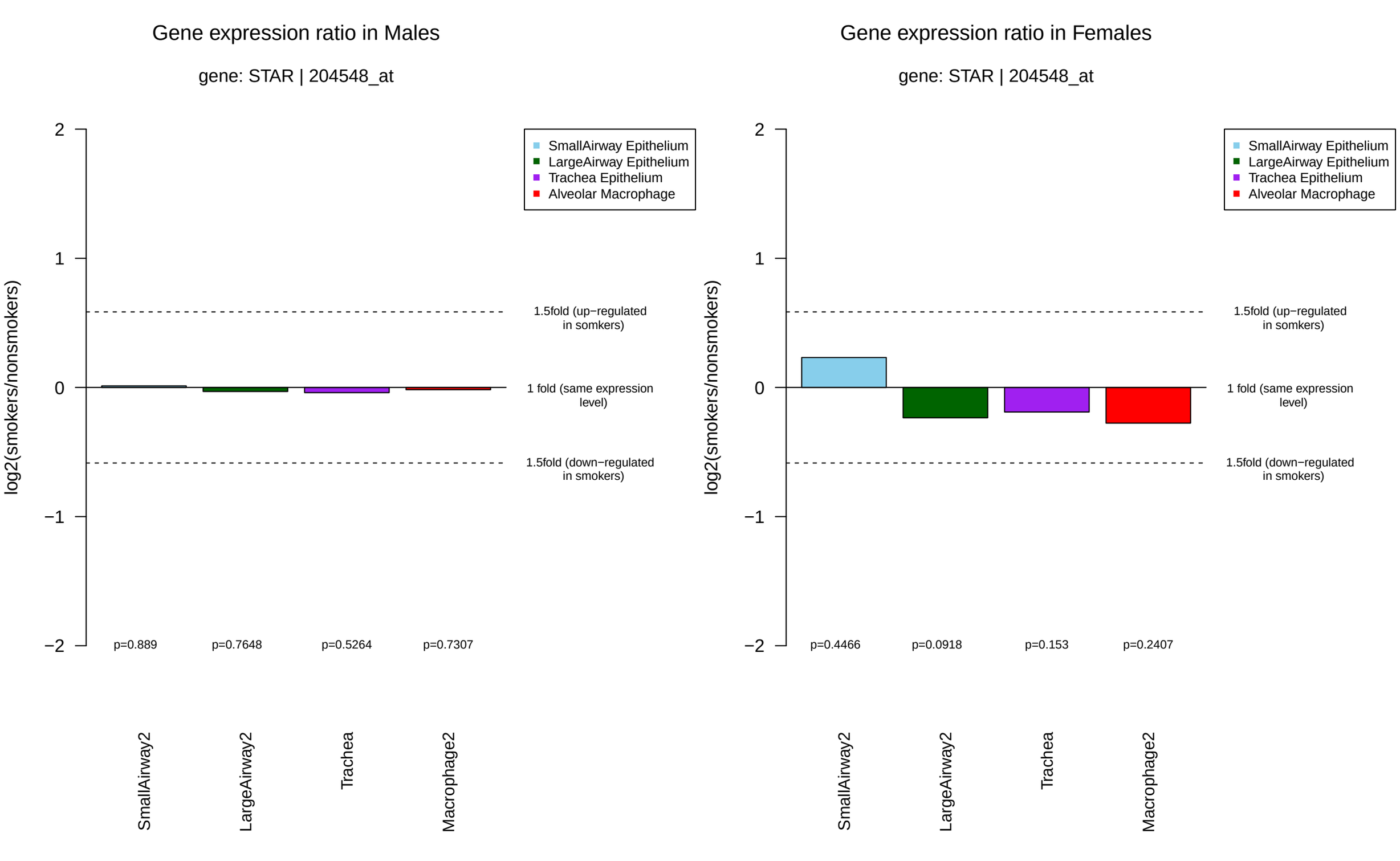
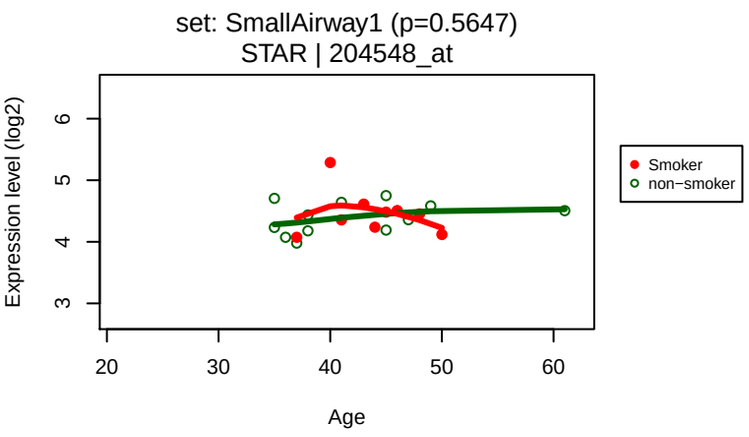
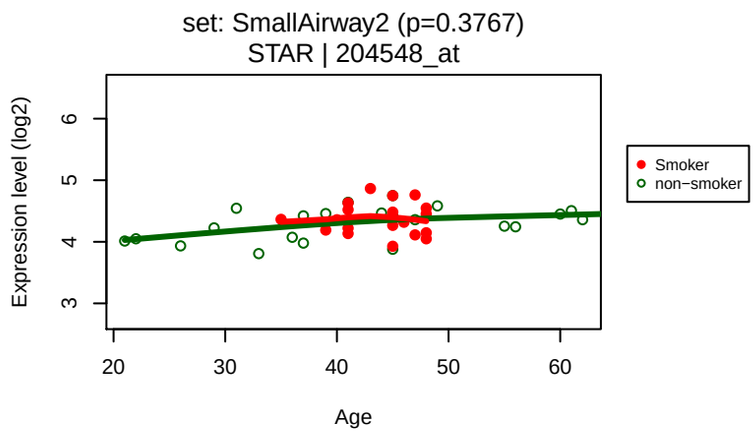
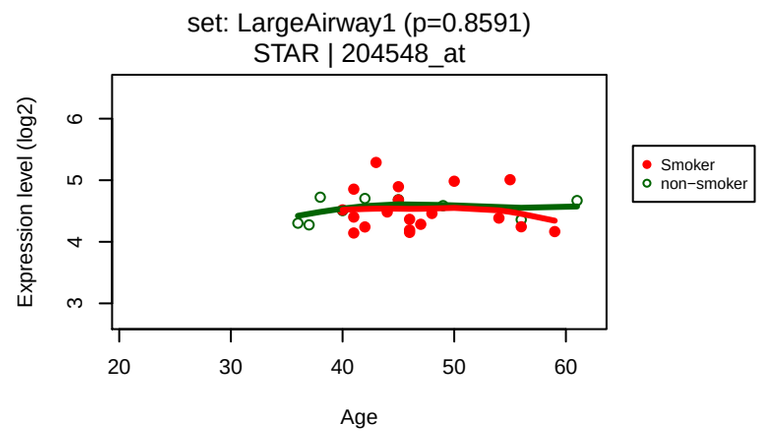

Gene expression of smokers and nonsmokers
gene: STAR | 204548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5647
p=0.3767
p=0.7509
p=0.8591
p=0.265
p=0.1967
p=0.7926
p=0.1283
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

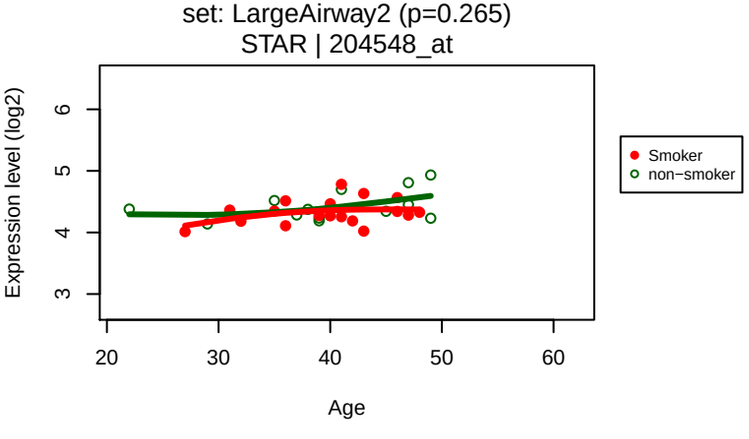

Gene expression ratio (smokers/nonsmokers)
gene: STAR | 204548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5647
p=0.3767
p=0.7509
p=0.8591
p=0.265
p=0.1967
p=0.7926
p=0.1283
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

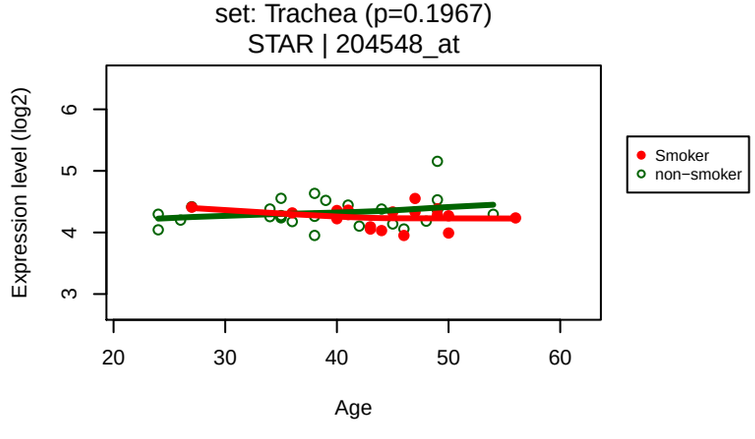

Gene expression ratio in Males
gene: STAR | 204548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.889
p=0.7648
p=0.5264
p=0.7307
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAR | 204548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4466
p=0.0918
p=0.153
p=0.2407
SmallAirway2
LargeAirway2
Trachea
Macrophage2

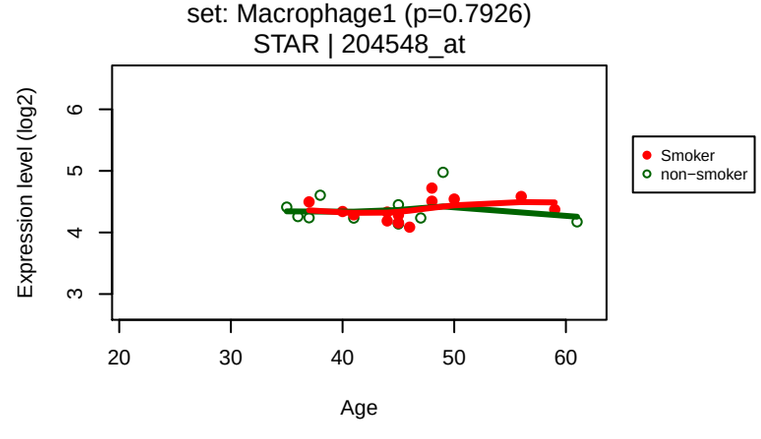

set: SmallAirway1 (p=0.5647)
STAR | 204548_at
Expression level (log2)
Age
Smoker
non−smoker

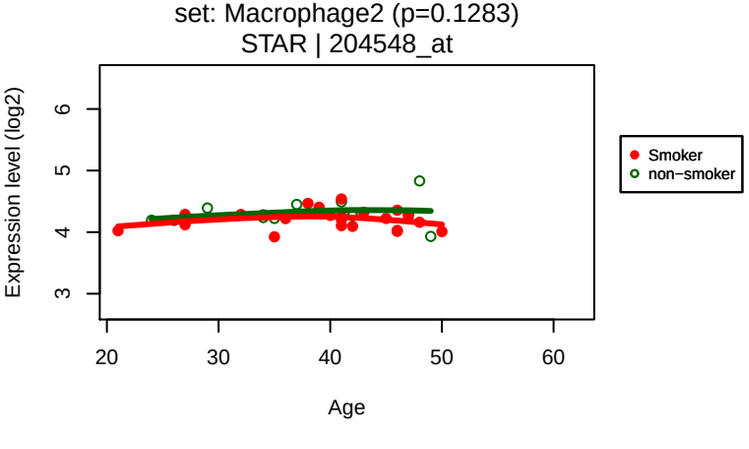

set: SmallAirway2 (p=0.3767)
STAR | 204548_at
Expression level (log2)
Age
Smoker
non−smoker

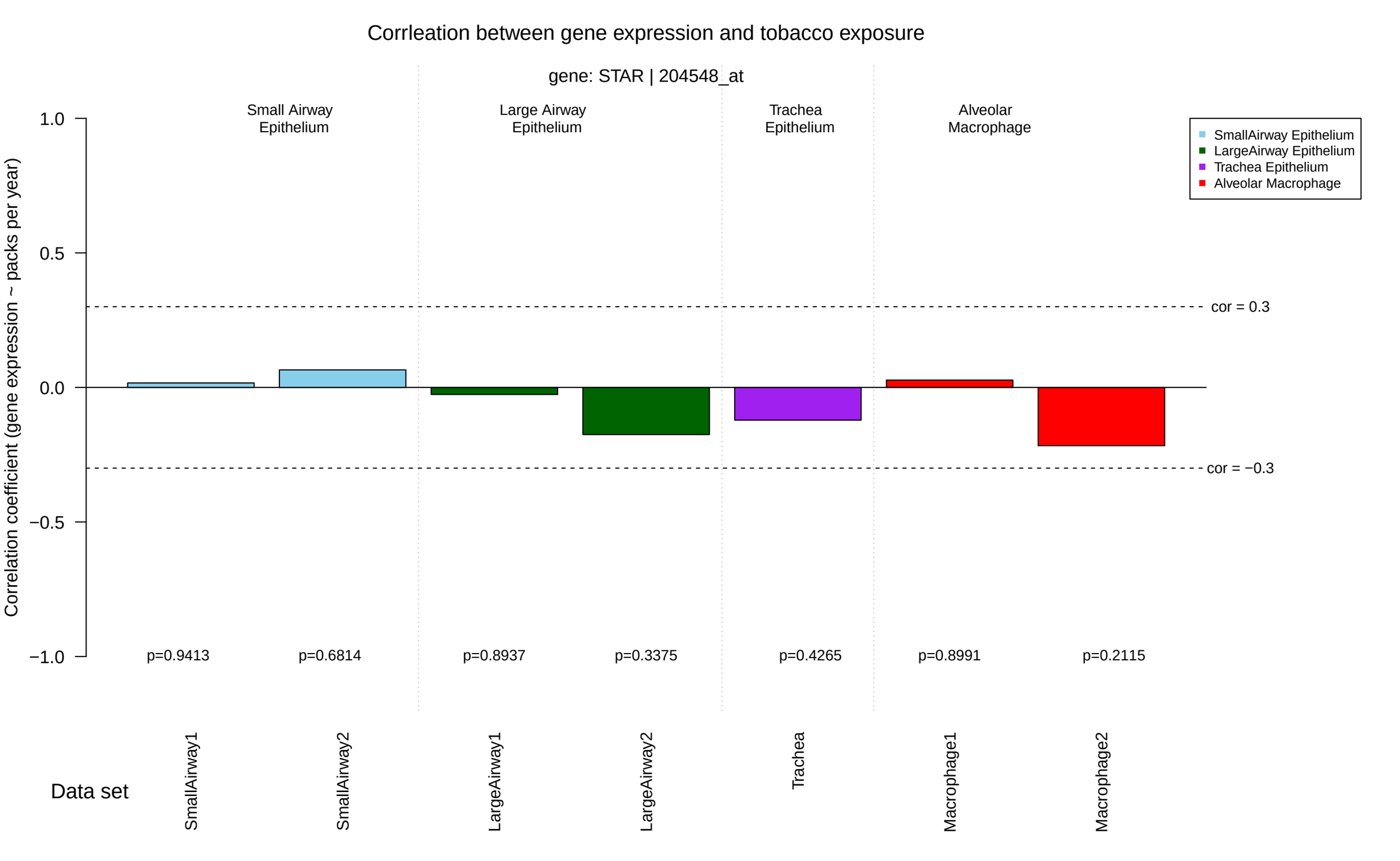

set: LargeAirway1 (p=0.8591)
STAR | 204548_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.265)
STAR | 204548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1967)
STAR | 204548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7926)
STAR | 204548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1283)
STAR | 204548_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAR | 204548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9413
p=0.6814
p=0.8937
p=0.3375
p=0.4265
p=0.8991
p=0.2115
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2